

Advanced Lab I, Fall 2026

Meeting Days: 8/26/2026 - 12/17/2026

Room: Peterson Hall 122

Times: PHY 236 01: T 1:00 PM-3:50 PM

PHY 236 02: R 10:00 AM-12:50 PM

PHY 236 03: Self-paced

Instructor Name: Xiang Li

Contact Information: xili@coe.edu Office: Peterson 141

Student Hours: Mon/Tue/Fri, morning until noon, or by appointment:

<https://calendar.app.google/22xRLNnu5xH8MQQw8>

Department Chair: Firdevs Duru (FDuru@coe.edu)

Course Description: PHY-236 Advanced Laboratory 1 (WE) Introduces laboratory measurement techniques as applied to modern physics experiments. Experiments from optics and atomic physics. A writing intensive course with laboratory and computer design projects based on applications of modern physics and atomic physics. Corequisite: Modern Physics (PHY-235). (0.5 course credit).

Measurable Learning Objectives: After successfully completing the course, the students are expected to understand the fundamental laws of quantum phenomena, such as wave behavior of particles and particle behavior of waves. Students will learn to write a scientific manuscript by preparing the weekly laboratory reports in the format of a peer-reviewed scientific paper. By working together with their laboratory partners, students will learn scientific collaboration. Please note that this course is Writing Emphasis (WE).

Required Texts and/or Course Materials: None. We will provide printed experiment guides for free.

Course Policies:

Schedule of major due dates: Assignments will typically be due 1 week post due unless specified. Due date will typically on a Friday night (9:00 pm). We will cover atomic spectra, radioactivity, determining the ratio of e/m , blackbody radiation, the photoelectric effect, X-ray diffraction, quantum analogs, spin and magnetic dipoles, electron diffraction, and gamma-ray spectroscopy. We will carry out 10 laboratory experiments, including the independent lab. Some weeks there will be no labs due to instructor absence or college breaks, but the room will be available for student use (to catch up on any late experiments or reports).

Method for determining final grade: There will be no curve. You get the grade you earn. If everyone gets an 'A', great! Each lab report (one is submitted for each group) will be worth:

Minor Lab Reports and assignments: 60%

Major Lab Reports: 30%

Independent experiment: 10%

Participation expectations, if applicable: Please make sure that you hand in assignments like homework or lab reports. PLEASE BE ON TIME COMING TO THE LAB to avoid late starts and disruptions.

Late work policy, if applicable: Late submission will get a point degradation penalty based on how late the submission is.

Acceptable methods for submitting work: Assignments and lab reports will be submitted to the Course Moodle page.

Course management system /website, if applicable: Instructions, templates, and reference guides will be available on the Course Moodle page.

Information about Final Exam and course meetings through the end of the term: There'll be no final exam for this course.

Any additional information or class policies on ChatGPT/AI/generative technology

If you are unsure whether a particular use is acceptable, ask me before you submit.

General rule of thumb: AI tools are roller skates, not crutches. AI tools (ChatGPT, Gemini, Claude, Copilot, and similar) may be used as long as they support your learning and do not replace your independent thought or effort.

Allowed: using AI to explain a derivation you are stuck on, generate boilerplate or plotting code, suggest debugging strategies, or brainstorm project ideas.

Prohibited: submitting AI-generated content that you cannot explain, derive, or defend; submitting AI output as if it were your own work; using AI during in-class quizzes or exams.

Required documentation. Any assignment where you used AI must include an AI Appendix that shows you vetted and validated the output:

Code — wrap AI-generated blocks in comments:

```
### Begin AI Code ###
```

```
...
```

```
### End AI Code ###
```

Images — caption clearly, e.g. [Image generated by (tool name)].

Prompts — include the prompts that materially shaped your submission, as a log at the end.

Validation — one or two sentences on how you checked the output was correct (dimensional analysis, a limiting case, a hand calculation, a numerical cross-check).

College-wide Policies:

Mission Statement:

Coe College is a national, residential liberal arts college offering a broad array of programs in the arts, sciences, and professions. Our mission is to prepare students for meaningful lives and fulfilling careers in a diverse, interconnected world. Coe's success will be judged by the success of our graduates.

Academic Integrity

At Coe College, we expect academic integrity of all members of our community. Academic integrity assumes honesty about the nature of one's work in all situations. Such honesty is at the heart of the educational enterprise and is a precondition for intellectual growth. Academic dishonesty is the willful attempt to misrepresent one's work, cheat, plagiarize, or impede other students' academic progress. Academic dishonesty interferes with the mission of the College and will be treated with the utmost seriousness as a violation of community standards. Please refer to the Coe College Academic Catalog for complete information regarding Academic Integrity: www.coe.edu/academics/academic-resources/provosts-office/academic-integrity-policy

[A Course-specific statement regarding the use of AI in the course - see below for optional language]

Respectful Communication

Coe College's effective teaching and learning environment is built on respectful communication. This includes avoiding harmful language or behaviors and approaching any mistakes as an opportunity to learn how to do better together.

FERPA

Students should be aware of their rights regarding the privacy of their educational records. Detailed information about your rights can be found under the FERPA (Family Educational Rights and Privacy Act of 1974) section in the Academic Catalog and online here: <https://www.coe.edu/academics/academic-resources/registrar/ferpa>

In line with FERPA restrictions, students should be aware that an instructor cannot publicly post grades by student name, institutional student identification number, or social security number without first having obtained students' written permission.

The Definition of a Course Credit & Expected Workload

One course credit at Coe College constitutes 180 hours' worth of student work

over the course of the term. "The Department of Education has defined one hour to be 50 minutes, so 150 60-minute hours is equivalent to 180 50-minute hours." This figure includes both the time spent in class and out of class completing course work. In other words, students are expected to devote a considerable amount of time outside of class to this course. For courses that meet in a standard MWF or T-Th slot, students should be expected to work seven hours a week outside of the three hours in class.

Requesting Academic Accommodations

Coe College is committed to creating a learning environment that meets the needs of its diverse student body. If you anticipate or experience any barriers to learning in this course, please visit my office hours to discuss these concerns with me.

If you have a medical, physical, psychological, or learning need documented by a healthcare provider, or experience an unforeseen obstacle during the semester, you can make an appointment with Melanie Ostmo, Accessibility Services Coordinator, to request an official accommodation under the Americans with Disabilities Act. You can find more information about the Accessibility Services Office by visiting the [Learning Commons](#) website or by emailing accessibility@coe.edu. If you have already been approved for accommodations through Accessibility Services, please meet with me so we can develop a plan for your success together.

Reporting of Sexual Assault or Misconduct

As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I also have a mandatory reporting responsibility related to my role as a faculty member. It is my goal that you feel able to share information related to your life experiences in classroom discussions, in your written work, and in any one-on-one meetings. I will keep the information you share with me private to the greatest extent possible. However, I am required to share information regarding sexual misconduct or students who may be in danger to themselves or to others. Students may speak to someone confidentially by contacting Student Life at 319-399-8843 or emailing titleix@coe.edu, Safety, and Security at 319-399-8888, Emily Barnard (college counselor) at 319-399-8843, or visit Coe's [Title IX website](#) for more information.

Optional Language

Statement on Generative Technology

Use of generative/assistive writing technology (such as ChatGPT or similar) without express permission from your instructor is a form of academic dishonesty. If specific parameters of use are not outlined by an instructor, students should assume use of said technology is prohibited, and that unauthorized use of it will be subject to disciplinary action. Coe College faculty may set parameters for when and how the use of such technology is permissible, as well as how students should document their use of it.